

Route Shield- A Crime Aware Smart Navigation System

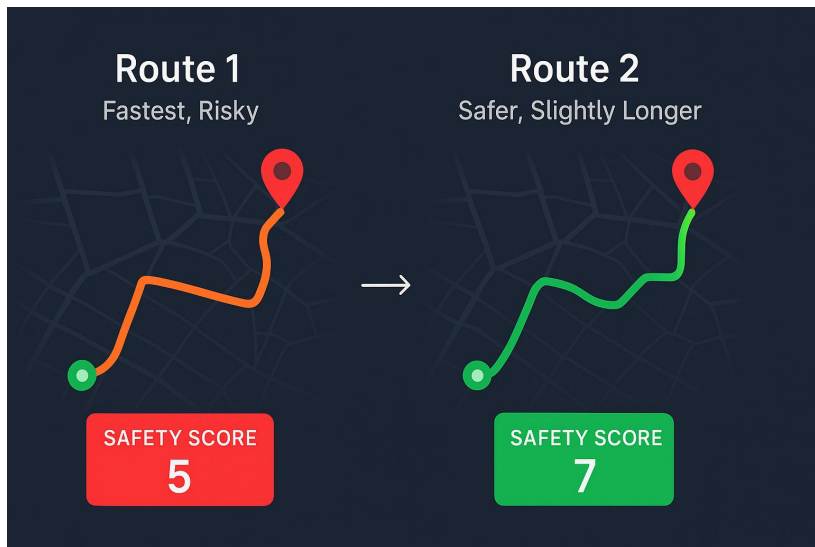
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What is Route Shield



Smart Navigation with Safety Focus: Finds not only the fastest routes but also the safest ones by integrating safety considerations into route planning.

Crime Data Integration: Combines crime data to ensure journeys are both efficient and secure.

LLM-Powered Crime Analysis: Analyze crime reports from Bangla newspapers, identifying crime types, time of occurrence, and locations to generate accurate insights.

Interactive Safety Visualization: Displays route-specific Safety Scores on an interactive map and highlights crime-prone areas using dynamic heatmaps for enhanced user awareness.

Motivation

Rising Urban Crime Challenges

Frequent incidents of mugging, theft, and harassment especially at night in Dhaka create growing safety concerns for daily commuters and travelers..

Lack of Localized Safety Intelligence

Existing navigation systems do not leverage crime news or real local incident data, leaving users unaware of route-specific safety risks.

Quality Function Deployment

Normal Requirements

Crime-Aware Route Planning: The system provides multiple routes between a source and destination while integrating crime data to assess safety risks along each path.

Safety Comparison & Scoring: Each route is assigned a clear Safety Score, allowing users to compare alternatives based on their preferred balance between safety and convenience.

Interactive Map Visualization: Routes and crime hotspots are displayed on an interactive map, helping users easily understand spatial risks and make informed navigation decisions.

Quality Function Deployment

Expected Requirements

Context-Aware Safety Scoring: Safety Scores are calculated using time-based crime patterns and weighted crime type classifications, ensuring route risk assessments reflect real-world safety conditions.

Insightful Crime Visualization: Precisely mapped crime data is presented through an interactive dashboard, allowing users to explore trends, patterns, and route-specific safety risks effectively.

Exciting Requirements

Personalized Safety Experience: Users can customize safety preferences by prioritizing specific crime types, allowing Safety Scores and route recommendations to align with individual risk tolerance and personal concerns.

Crime Trend Analysis: An interactive dashboard presents neighborhood-level crime trends, detailed incident breakdowns, and temporal patterns, enabling users to better understand and anticipate urban safety dynamics.

Workflow

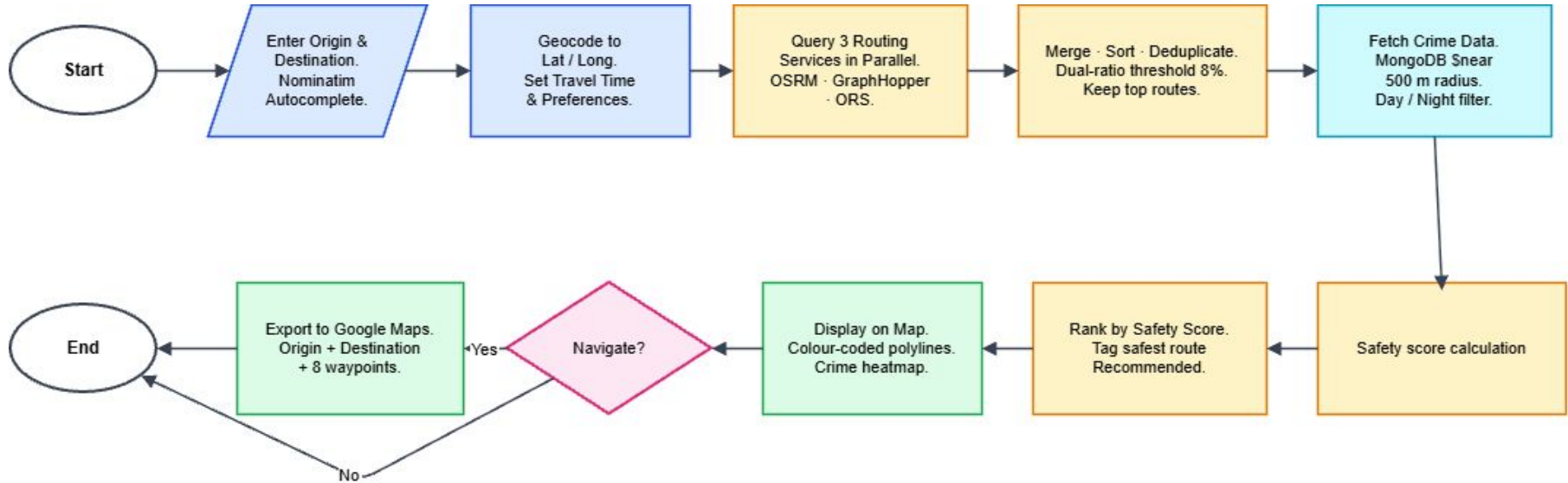
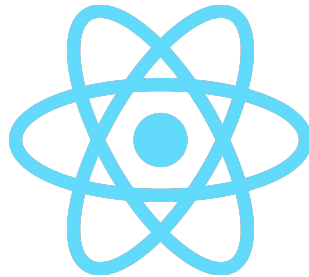
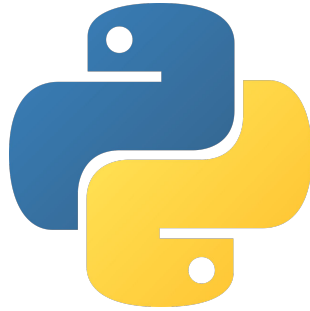


Figure 1 — Route Shield: Crime-Aware Navigation Workflow

Tools and Technologies



Project Demonstration

<https://route-shield-one.vercel.app/>

Test Case Check

T8

Route Comparison

View multiple route options with Safety

Clear display of all options; distance/score shown for



★ Recommended — Safest option

Route 1 ☒

Low Risk 11 crimes on route [See Crimes](#)

Safety Score 7.6/10

Safe to travel Safer than 100% of routes shown

Duration 19 mins Distance 18.1 km

This route is relatively safe with 11 crime(s) nearby. Be cautious in 1 area(s).

[Navigate via this Route in Google Maps](#)

Opens Google Maps with waypoints from this exact route

✓ Visible on map • Click to hide

Route 2 ☒

Low Risk 17 crimes on route [See Crimes](#)

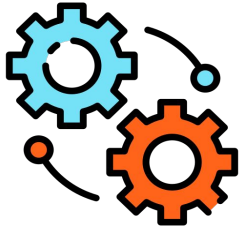
Safety Score 7.5/10

Safe to travel Safer than 50% of routes shown

Duration 19 mins Distance 22.9 km

This route is relatively safe with 17 crime(s) nearby. Be cautious

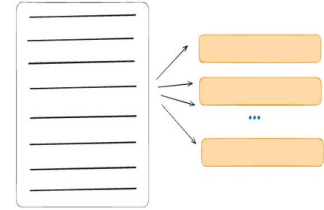
Challenges



Multi-Service Route Aggregation and
Deduplication



Personalized Multi-Factor Safety Scoring



Geocoding Ambiguity for Dhaka
Locations

Future Scope

Real-Time Crime Data Integration

Additional Travel Modes (Walking and
Public Transport)

User-Adjustable Proximity
Threshold and Data-Driven
Crime Weights



Thank You

